SUBSTACKS OF X CORRESPOND TO R-INVARIANT LOCALLY CLOSED SUBSPACES OF A PRESENTATION U

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2025-2039 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 2041-2063, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_stacks-properties.json (pb=1260, stmt 2025-2039, proof 2041-2063).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-immersion-into-presentation: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1896–1910. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-immersion-presentation: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1936–1948. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $[U/R] \to \mathcal{X}$ be a presentation of an algebraic stack. There is a canonical bijection*

$$\text{locally closed substacks } \mathcal{Z} \text{ of } \mathcal{X} \longrightarrow R\text{-invariant locally closed subspaces } Z \text{ of } U$$

which sends $\mathcal{Z}$ to $U \times_{\mathcal{X}} \mathcal{Z}$. Moreover, a morphism of algebraic stacks $f : \mathcal{Y} \to \mathcal{X}$ factors through $\mathcal{Z}$ if and only if $\mathcal{Y} \times_{\mathcal{X}} U \to U$ factors through Z. Similarly for closed substacks and open substacks.

Proof. By Lemmas lemma-immersion-into-presentation and lemma-immersion-presentation we find that the map is a bijection. If $\mathcal{Y} \to \mathcal{X}$ factors through $\mathcal{Z}$ then of course the base change $\mathcal{Y} \times_{\mathcal{X}} U \to U$ factors through Z. Converse, suppose that $\mathcal{Y} \to \mathcal{X}$ is a morphism such that $\mathcal{Y} \times_{\mathcal{X}} U \to U$ factors through Z. We will show that for

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every scheme T and morphism $T \to \mathcal{Y}$, given by an object y of the fibre category of $\mathcal{Y}$ over T, the object y is in fact in the fibre category of $\mathcal{Z}$ over T. Namely, the fibre product $T \times_{\mathcal{X}} U$ is an algebraic space and $T \times_{\mathcal{X}} U \to T$ is a surjective smooth morphism. Hence there is an fppf covering $\{T_i \to T\}$ such that $T_i \to T$ factors through $T \times_{\mathcal{X}} U \to T$ for all i. Then $T_i \to \mathcal{X}$ factors through $\mathcal{Y} \times_{\mathcal{X}} U$ and hence through $Z \subset U$. Thus $y|_{T_i}$ is an object of $\mathcal{Z}$ (as Z is the fibre product of U with $\mathcal{Z}$ over $\mathcal{X}$). Since $\mathcal{Z}$ is a strictly full substack, we conclude that y is an object of $\mathcal{Z}$ as desired. $\square$